

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (25 marks – approximately 45 minutes)

Savannah Pharmaceutical Products Limited ("SPPL") uses a process costing system and the first-in, first-out cost-flow assumption. SPPL produces a drug from two processes. 50,000 units were started in April with material costs of HK\$119,500 and conversion costs of HK\$105,380. 51,000 units were completed in April and transferred to Process 2. The following data relate to Process 1 for the month of April.

		Percentage of completion		Costs	
	Units	Direct materials	Conversion	Direct materials	Conversion
Work in process, 1 April	8,000	80%	60%	HK\$18,000	HK\$10,800
Work in process, 30 April	6,000	70%	45%		

A normal loss of 2,000 units occurred during the process. The normal loss was absorbed by both completed units and closing work in process, as both completed units and closing work in process have reached the inspection point detecting normal loss. All losses were scrapped with nil value.

Required:

- (a) Calculate the equivalent units for direct materials and conversion costs, and the cost per equivalent unit for direct materials and conversion costs for the month of April.
(10 marks)
- (b) Prepare the Process 1 account for the month of April.
(9 marks)
- (c) Explain normal losses, abnormal losses and abnormal gains.
(6 marks)

Question 2 (18 marks – approximately 32 minutes)

Gwenda Company Limited ("GCL") receives a special order for 20,000 units of one of its former products, Ruby, at HK\$70 each from an overseas company.

The order requires 12,000 kg of material A. There are 2,000 kg of material A in inventory which are not required for normal production. These materials were purchased at a cost of HK\$21,000. If these materials are not used in this order, they could be sold for HK\$10 per kg. The current replacement cost of material A is HK\$12 per kg.

The order also requires 21,000 kg of material B. There are 1,800 kg of material B in inventory which were purchased at a cost of HK\$32,400. Material B is used regularly by GCL in other production. Currently, the price for material B has increased by HK\$2 per kg.

The order requires 1,500 hours of skilled labour. There is a limited supply of skilled labour, and all the available skilled labour is fully engaged in the production of another product, Mazarine. Mazarine is sold at HK\$500 per unit and its production costs are as follows:

	HK\$ per unit
Direct material C	120
Skilled labour (2 hours)	160
Variable production overheads	40
Fixed production overheads	<u>60</u>
	<u>380</u>

Variable production overheads of HK\$11 will be incurred per unit of Ruby. Additional fixed production overheads of HK\$100,000 will be incurred if the special order is taken and the fixed production overheads will be absorbed at HK\$8 per direct labour hour. An additional selling and distribution overhead of HK\$60,000 will also be incurred for the order.

An income tax of 2% on the sales value of the order will be payable to the tax authority of the overseas customer.

Required:

- (a) Calculate the relevant costs of direct materials, direct labour, variable production overheads, fixed production overheads, selling and distribution overhead, income tax and opportunity cost, if any, for the special order. Justify whether the special order should be taken, based on financial consideration only.

(12 marks)

- (b) Define fixed costs and justify whether fixed costs are relevant in short-term decision-making.

(6 marks)

Question 3 (22 marks – approximately 40 minutes)

Jeanette Technology Limited ("JTL") is a leading manufacturer for micro automatic guided vehicles. JTL uses a standard costing system, and the standard cost information for its single product for the year is as follows:

	HK\$
Direct materials (10 kg at HK\$15 per kg)	150
Direct labour (2 hours at HK\$60 per hour)	120
Variable production overheads (2.5 machine hours at HK\$20 per hour)	50
Fixed production overheads	<u>80</u>
Standard production cost per unit	<u>400</u>

Planned and actual production for the year was 30,000 units. There was no opening inventory of direct materials and there was a 40,000 kg closing inventory of direct materials. JTL calculates the price variance of direct materials at the time of purchase.

The following variances for the year are provided:

Direct materials price variance	HK\$320,000 (Adverse)
Direct materials quantity variance	HK\$300,000 (Favourable)
Direct labour rate variance	HK\$122,000 (Favourable)
Direct labour efficiency variance	HK\$60,000 (Adverse)
Variable production overheads spending variance	Nil
Variable production overheads efficiency variance	HK\$100,000 (Favourable)
Fixed production overheads spending variance	HK\$500,000 (Adverse)

Required:

(a) Calculate the following:

- (i) Direct materials used in kg;
- (ii) Direct materials purchased in kg;
- (iii) Direct materials purchased in dollars;
- (iv) Actual direct labour cost;
- (v) Actual direct labour hours used;
- (vi) Actual direct labour rate per hour;
- (vii) Actual variable production overheads;
- (viii) Actual machine hours used; and
- (ix) Actual fixed production overheads.

(14 marks)

(b) Describe ideal standards and currently attainable standards. Explain which types of standards are used in setting standard costs for most companies.

(8 marks)

Question 4 (15 marks – approximately 27 minutes)

Veronica Beauty Group ("VBG") is a leading beauty and wellness services provider in Hong Kong and mainland China that targets high-end customers. VBG has two business segments: Beauty Division and Spa Division. The financial results for last year of the two divisions are as follows:

	Beauty Division HK\$	Spa Division HK\$
Operating assets	5,000,000	12,000,000
Operating profit	550,000	1,560,000

Required:

- (a) Calculate the return on investment in percentage and the residual income for each division. The required rate of return of VBG is 12%.

(6 marks)

- (b) VBG is considering using a Balanced Scorecard to evaluate its performance. The management accountant has determined that VBG will use the net profit and the residual income as two measures under the financial perspective of the Balanced Scorecard. List the other THREE perspectives in the Balanced Scorecard and identify TWO measures for VBG for each of the THREE perspectives.

(9 marks)

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